

Information Retrieval

Assignment A1

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Architecture & Workflow

Environment: Python-based, entirely command-line operated.



Core Logic

`main.py`

Handles inverted index construction, term-weighting, and vector space similarity computations.



Data Parsing

`parser.py`

Parses TREC-formatted topics and SGML-based document collections.



Utilities

`process_wordnet_cs.py`

Additional utilities for parsing Wordnet data.

Key Features & Capabilities



Preprocessing

Includes lemmatization via the MorphoDiTa tagger, case folding, POS-based stopping, and number normalization.



Weighting

Supports various schemes (e.g., nnn, ntc, ltn, ltc) for both queries and documents.



Query Expansion

Thesaurus-based expansion using NLTK WordNet for English or WordNet 1.9 PDT for Czech.



Performance

Features multiprocessing to handle large collections efficiently.

Run-0 and Run-1 Settings

Settings for Run-0 (Baseline)

- **Tokenization:** whitespace + punctuation
- **Class equivalence:** no
- **Removing stopwords:** no
- **Query construction:** all words from "title"
- **Term weighting:** natural
- **Doc freq weighting:** none
- **Vector normalization:** cosine
- **Similarity:** cosine
- **Relevance feedback:** none
- **Query expansion:** none

Settings for Run-1 (Optimized)

- **Tokenization:** whitespace + punctuation
- **Class equivalence:** case folding, lemmatization
- **Removing stopwords:** none
- **Query construction:** Topic titles only
- **Term weighting:** Logarithmic (l)
- **Doc freq weighting:** IDF (t) docs; None (n) queries
- **Vector normalization:** None (n) docs; Cosine (c) queries
- **Similarity:** Cosine similarity
- **Relevance feedback:** none
- **Query expansion:** Thesaurus-based (scaling 0.2)

Evolution Run-0 → Run-1

Change	English (MAP)	Czech (MAP)
Run-0	0.0411	0.0596
Change weighting to ltn.lnc	0.2239	0.1626
Add lemmatization	0.2527	0.3224
Add Query expansion	0.2807	0.3236

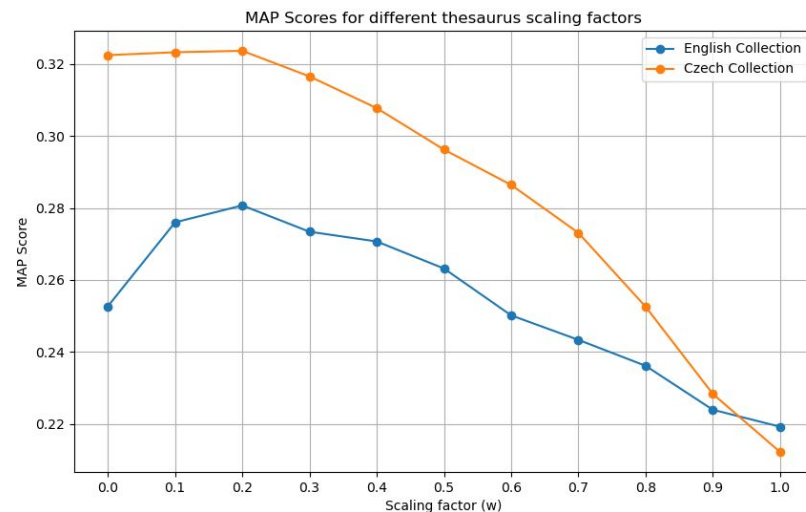
Experiments & Parametric Optimization

Methodology: Conducted 90+ experiments to determine optimal retrieval parameters.

Preprocessing Impact

Approach	English	Czech
POS-based stopping	0.1865	0.1624
Lemmatization	0.2527	0.3224
Combined	0.2498	0.3177

Weighting Scheme Optimization



Final Performance Results

Run-0 (Baseline)

ENGLISH RESULTS

MAP 0.0411 | P@5 0.0649

CZECH RESULTS

MAP 0.0596 | P@5 0.0880

Run-1 (Constrained)

ENGLISH RESULTS

MAP 0.2807 | P@5 0.4400

CZECH RESULTS

MAP 0.3236 | P@5 0.4320

*MAP: Mean Average Precision | P@5: Precision at rank 5. Optimization in Run-1 shows significant improvement across both languages.

Conclusion & Questions



Thank you for your attention.